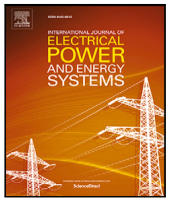
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Recovering fixed costs in the presence of prosumers

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ABSTRACT

The existence of prosumers who concurrently produce and consume electricity with distributed renewable resources such as photovoltaic generation has been viewed as an effective way of fortifying grid resilience and enhancing sustainability. However, prosumers' self-consumption is impacted by how surplus electricity is priced when feeding it into the grid. From the perspective of fixed cost recovery, this study compares market outcomes when their transactions with the grid are subject to net-metering and net-billing policies considering both the wholesale and retail markets. Under the net-billing policy, prosumers may face a power price lower than the retail rate or the wholesale price when selling their surplus energy into the grid. Compared to the benchmark case, where prosumers are subject to the same price signals as conventional consumers and producers, prosumers would over-consume under the net-billing case. The net-billing policy converges to the net-metering policy when prosumers purchase power from the grid. The analysis also finds that the net-metering policy can result in more wealth transfer from conventional producers and consumers to prosumers when compared to the net-billing policy.

1. Introduction

Driven by a need for decarbonization, sustainability, and resilience, the energy sector in the U.S. and elsewhere has witnessed a remarkable move toward advancing and deploying distributed energy resources (DERs). The paradigm in the energy marketplace has shifted toward a more demand-side engagement to harness flexible DERs. This inevitably challenges the conventional, top-down power grid architecture based on the supply side, and unleashes opportunities for a new market design for the power sector [1]. In the meantime, the U.S. Federal Energy and Regulatory Commissions (FERC) paved the way by issuing FERC Order 745 in 2011, ensuring that demand response providers must be compensated for reducing electricity load at the same rates as if they met that demand with generated electricity. The U.S. Supreme Court ruling on the order in 2016 further vindicates the agency's role in regulating demand-side resources in the wholesale market [2].

More recently, the FERC issued a landmark order, FERC Order 2222, which removes barriers to the entry of emerging technologies, thereby breaking new ground toward creating the grid of the future [3]. More specifically, the ruling allows the coupling of multiple DERs with different sizes and diverse technologies to participate in the regional organized wholesale capacity, energy, and ancillary services markets alongside traditional resources. In fact, there are already some activities in the marketplace in response to or in anticipation of the order. Examples include the recently announced plan by OhmConnect to link homes scattered in California to form a 550 MW of distributed energy resources [4]. The project will allow the homeowners to "value-stack" their roof-top solar panels by providing energy and various types of services to the grid. Another example is a company that acts as an aggregator managing more than 280 commercial, industrial, and residential customers under the California Independent System Operator (CAISO) to provide energy and frequency regulation among other services [5].

The development is certainly welcomed by prosumers or community choice aggregators who actively engage in the energy market. However, it also creates concerns. In particular, the question of how to price the energy produced by prosumers, also related to the recovery of fixed costs for utilities, is subject to debate [6]. Not surprisingly, prosumers are self-served for their own interests, and their behavior resulting from optimizing their private benefit is unlikely to be in the best interests of the energy market as a whole. Furthermore, the fact that FERC Order 745 focuses mainly on demand response resources implies that this surplus energy may not necessarily be subject to the same

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treatment even if it resulted from the efforts of prosumers to conserve energy demand.[2] In this case, the issue is the counterfactual as the prosumers' load in the absence of an opportunity to sell back to the grid has a deep root in economics and has been subject to contentious debates in evaluating demand response programs.

Efficient integration of DERs owned by prosumers undoubtedly requires not only new technologies, but also many regulatory and market changes. Changes to tariff designs, particularly, pricing rules for energy produced by DERs, has received some attention in the literature. Prosumers, whose costs are mostly recovered by volumetric fees, may end up paying less for the infrastructure than conventional consumers by relying less on the network, thereby adversely affecting the financial viability of utilities [10,11].[3] The situation is also dubbed "revenue erosion." In this situation, utilities are forced to raise the grid tariff to compensate for the revenue deficiency, further exacerbating the situation and leading to a downward spiral [12–15]. There is some empirical evidence documenting this phenomenon. For instance, using data from three public utilities in California, Wolak [16] finds that two thirds of increases in residential distribution prices can be attributed to growing solar capacity. Eid et al. [17] consider distributed generators that can offset their own power consumption with local generation under a net-metering policy. Using consumption and production data on low-voltage photovoltaic (PV) prosumers in Spain, they find that utilities reduced their income, which pushed up network tariffs for cost recovery.

More recent studies on prosumers focus on cost allocations for transmission or distribution-related assets under different pricing schemes. Using stylized models, Gautier et al. [18] conclude that net metering decreases the payment of prosumers, which is cross-subsidized by the higher bills of conventional consumers. The system may lead to too many prosumers and provide no incentive to synchronize local production and consumption. Brown and Sappington [19] suggest that a net-metering scheme can be suboptimal depending on the levels of the fixed costs of centralized energy production and the network management costs. On the other hand, Schittekatte et al. [20] model the fixed-cost recovery problem as a non-cooperative game among prosumers and consumers. They argue that the net-purchasing system, in which different prices are set on the electricity imported from or exported to the grid, may be a better option in a future scenario of low technology costs.

The current study extends the previous work [21] to examine the effects of prosumers on the short-run market equilibria in the electric power sector considering the link between the wholesale and retail markets. Thus, all the capital costs are sunk, thereby not affecting operations decisions. Different from [21], which examines the extent of prosumers' market power potential in the wholesale market, this paper considers a situation in which the utility's fixed cost of the network infrastructure needs to be recovered under different pricing schemes. Under a net-billing system, two meters are installed, recording two quantities, i.e., power withdrawn from and power injected into the grid, where withdrawal and injection can be subject to difference prices. In contrast, a single meter can be installed under a net-metering system to record both transactions, subjecting them to the same price. These two policies are then compared to the benchmark case in which prosumers are treated in the same way as conventional consumers (producers) when consuming (selling) power. This study is related to [18], but is particularly different in that the current work considers price-responsive prosumer demand for electricity, transmission network, and social welfare under different pricing schemes for prosumers, which are crucial in evaluating outcomes of the electricity market. Table 1 compares the current work and other recent studies on prosumers with similar topics to see whether key elements for analysis are incorporated.

The analysis focuses on the transmission-related costs, but it can be generalized to other situation as long as there is a fixed cost that needs to be recovered and collected through retail rates, such as costs to meet state renewable energy mandate.[4] While each prosumer may be relatively small, possessing limited ability to engage in the bulk energy market, this study assumes that an entity aggregates a large number of prosumers, both residential and industrial customers, and participates in the bulk energy market on their behalf, consistent with FERC Order 2222. Thus, analysis is more applicable to the situation facing the U.S. electric power industry. Prosumers are endowed with renewables and assumed to decide the amounts of self-consumption and energy to sell into or buy from the bulk energy market to maximize their net private benefit, which is characterized by their "retail" inverse demand function. The independent system operator (ISO) maximizes the social surplus based on the "wholesale" inverse demand while treating the sales or purchases by prosumers as fixed variables.

The current work contributes to (1) advancing modeling approach by explicitly considering the linkage between the retail and wholesale markets in which prosumers actively participate, (2) enhancing the community's understanding of the ongoing debate about how to price prosumers' energy output, particularly highlighting the intrinsic relations between the amount of renewables, size of prosumers as well as their joint effect on the market outcomes, and finally, (3) providing an evidence-based policy prescription to address the challenges facing the power sector to recover fixed costs of the network in the presence of prosumers.

The remainder of this paper is organized as follows. Section 2 formulates the simulation models. Section 3 analytically compares the net-metering and benchmark cases. A numerical case study is presented in Section 4. Finally, concluding remarks are provided in Section 5.

2. Models

This section proceeds as follows. First, it gives the notation used in the model, followed by the optimization problem of each entity in the market, that is, prosumers and the ISO. Second, it derives the Karush–Kuhn–Tucker (KKT) conditions for each optimization problem. Third, the collection of KKT conditions together with the condition for fixed cost recovery defines a market equilibrium problem, which can then be solved using complementarity solvers, such as PATH.

2.1. Notation

(1) Indices and Sets

$i \in \mathcal{I}$	Nodes.
$k \in \mathcal{K}$	Transmission lines.
$h \in \mathcal{H}_i$	Generating units at node i.

[2] For example, the recent decision by the Utah Public Service Commission (PSC) to slash solar export rates associated with Rocky Mountain Power has cast doubt about the future of residential solar in the state [7]. Recently, the policy debate on the provision for NEM (Net-Metering) 3.0 by the California Public Utility Commission (CPUC) is a testament to the challenges and concerns facing the electricity sector [8]. The most contentious issue concentrates on CPUC's intention to charge all customers a monthly grid participation fee of $8/kW based on their solar capacity; this, argued by some solar proponents, can effectively falter renewables development by reducing the value of new solar system by 50%–75% [9].

[3] Two distinct types of fixed costs need to be recovered in the power sector. One refers to capital costs of building power plants, which generators recover from the wholesale market when serving as inframarginal units. The second type relates to, for example, investment in transmission network, which is the focus of the paper. In perfectly competitive markets, "producer surplus" of generators is expected to cover the former type of fixed cost.

[4] At least in California, the PUC has included a torrent of policy goals into electricity retail rate making, e.g., subsidies for rooftop solar and electrical vehicle charging stations, discount for low-income customers, energy efficiency programs, school water quality, and more recently, wildfire mitigation and compensation [22].

Table 1
Comparison of recent studies on prosumers.

Studies\Key elements	Price-responsive demand	Network	Network-cost recovery	Social welfare[a]
Gautier et al. [18]	×	×	∘	×
Brown and Sappington [19]	∘	×	∘	∘
Schittekatte et al. [20]	×	×	∘	×
Ramyar et al. [21]	∘	∘	×	×
This study	∘	∘	∘	∘

[a]Social welfare under different pricing schemes.

(2) Parameters

K_i	Renewable output produced by prosumers at node i (MW).
G_i	Production capacity of prosumers' dispatchable unit at node i (MW).
α_i	Fraction of prosumers in node i (unitless).
P_i^0, Q_i^0	Vertical and horizontal intercepts of retail inverse demand function at node i, $p_i^r(d_i + l_i) = P_i^0 - (P_i^0/Q_i^0)(d_i + l_i)$ ($/MWh, MWh).
X_{ih}	Production capacity for conventional generation unit h at node i (MW).
F	Amortized fixed cost of transmission owners ($).
R	Ratio of τ^s to τ^b under net billing (unitless).
$PTDF_{ki}$	Power transfer distribution factor for a unit of power transferred from the hub to node i through line k (unitless).
T_k	Thermal limit for line k (MW).

(3) Primal Variables

z_i	Purchases (−) or sales (+) by prosumers at node i under net metering (MWh).
z_i^b, z_i^s	Purchases and sales by prosumers at node i under net billing (MWh).
τ	Transmission charge for purchases or subsidy for sales under net metering ($/MWh).
τ^b, τ^s	Transmission charge for purchases and subsidy/tax for sales under net billing ($/MWh).
l_i	Prosumers' demand at node i (MWh).
g_i	Energy produced by prosumers' dispatchable unit at node i (MWh).
x_{ih}	Energy generated by conventional generation unit h at node i (MWh).
d_i	Conventional consumer's demand at node i (MWh).
y_i	Energy injection/withdrawal at node i (MWh).

(4) Dual Variables

δ_i	Dual variable for prosumers' energy balance at node i ($/MWh).
κ_i	Dual variable for prosumers' dispatchable generation capacity at node i ($/MWh).
ρ_{ih}	Dual variable for capacity constraint of generation unit h at node i ($/MWh).
θ	Dual variable for total supply and demand balance ($/MWh).
λ_k^+, λ_k^-	Dual variables for limit of line k ($/MWh).
p_i	Dual variable for supply and demand balance at node i (wholesale power price at node i) ($/MWh).

2.2. Type of consumers

All consumers are grouped into two types, i.e., conventional consumers and prosumers, whose marginal benefit functions in node i, p_i^{con} and p_i^{pro}, are respectively represented by the following linear inverse demand functions at the retail level:

$$p_i^{con}(d_i) = P_i^0 - \frac{P_i^0}{(1-\alpha_i)Q_i^0} d_i \tag{1}$$

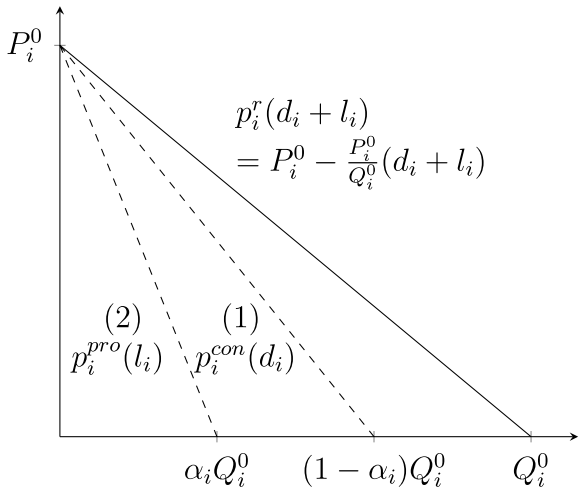


Fig. 1. An illustration of retail demand curves.

$$p_i^{pro}(l_i) = P_i^0 - \frac{P_i^0}{\alpha_i Q_i^0} l_i \tag{2}$$

where P_i^0 and Q_i^0 respectively represent the vertical and horizontal intercepts of the "horizontally aggregated" retail inverse demand function, $p_i^r(d_i + l_i) = P_i^0 - \frac{P_i^0}{Q_i^0}(d_i + l_i)$, as illustrated in Fig. 1.[5] The quantities demanded by conventional consumers and prosumers are denoted by d_i and l_i, respectively. The parameter α_i is the fraction of prosumers at node i. Note that while α_i varies between 0 and 1, the aggregated demand does not change. Let τ or τ^b denote a volumetric transmission charge based on any types of stranded or sunk costs that need to be recovered.[6] Then, $p_i = p_i^r - \tau$ or $p_i = p_i^r - \tau^b$ holds, where p_i is the wholesale power price at node i.

2.3. Prosumers

The model posits that prosumers maximize their profit by deciding (i) power to buy from or sell to the grid through node i, (ii) consumption l_i given renewable output $K_i > 0$, and (iii) generation g_i from the backup dispatchable technology, e.g., on-site generator or storage, with a cost $C_i^g(g_i)$.

Net Billing: Under a net-billing system, two meters are implemented that record two quantities: the power withdrawn from (z_i^b) and the power injected into (z_i^s) the grid through node i, respectively. The power injected into the grid is compensated for by $p_i + \tau^s$, where τ^s could be positive or negative. When $\tau^s > 0$, prosumers effectively receive a "subsidy" in addition to the wholesale price. In the case when $\tau^s < 0$, it means that prosumers are subject to a "tax" when

[5] See, for example, [23] for the details about the inverse demand function.
[6] τ or τ^b can also be regarded as the sum of volumetric transmission and distribution charges. In that case, the stranded costs are associated with both transmission and distribution networks.

selling power into the grid, contributing to fixed cost recovery.[7] On the other hand, the power withdrawn from the grid is priced at $p_i + \tau^b$, where τ^b is always positive as a usual transmission tariff for end-users. The optimization problem faced by prosumers at node i is written as follows.

$$\underset{z_i^s,\, z_i^b,\, l_i,\, g_i \geq 0}{\text{maximize}} \quad (p_i + \tau^s) z_i^s - \left(p_i + \tau^b\right) z_i^b + \int_0^{l_i} p_i^{pro}(m_i) dm_i - C_i^g(g_i) \tag{3a}$$

subject to

$$z_i^s - z_i^b + l_i - K_i - g_i \leq 0 \qquad (\delta_i), \tag{3b}$$

$$g_i \leq G_i \qquad (\kappa_i) \tag{3c}$$

The objective function (3a) consists of the revenue by selling z_i^s to the grid, the cost for purchasing z_i^b from the grid, the benefit of consuming power l_i, and the cost of backup generation g_i. Eq. (3b) makes sure that the net demand, $z_i^s - z_i^b + l_i$, is balanced with the sum of renewable and backup outputs.[8] Eq. (3c) limits the backup output to be less than their capacity G_i.

The corresponding KKT conditions of the prosumers' problem in node i under the net-billing system are collected for all nodes as in (4a)–(4f).

$$0 \leq z_i^s \perp p_i + \tau^s - \delta_i \leq 0, \forall i \tag{4a}$$

$$0 \leq z_i^b \perp -p_i - \tau^b + \delta_i \leq 0, \forall i \tag{4b}$$

$$0 \leq l_i \perp P_i^0 - \frac{P_i^0}{\alpha_i Q_i^0} l_i - \delta_i \leq 0, \forall i \tag{4c}$$

$$0 \leq g_i \perp -C_i^{g\prime}(g_i) + \delta_i - \kappa_i \leq 0, \forall i \tag{4d}$$

$$0 \leq \delta_i \perp z_i^s - z_i^b + l_i - K_i - g_i \leq 0, \forall i \tag{4e}$$

$$0 \leq \kappa_i \perp g_i - G_i \leq 0, \forall i \tag{4f}$$

Assuming interior solutions for the primal variables yields further interpretation of those conditions. When $z_i^s > 0$ and $l_i > 0$, $p_i^{pro}(l_i) = P_i^0 - \frac{P_i^0}{\alpha_i Q_i^0} l_i = p_i + \tau^s$ holds from Eqs. (2), (4a), and (4c). This implies that prosumers decide their sales z_i^s and consumption l_i in such a way that their marginal benefit is equal to the sum of the wholesale price p_i and the subsidy/tax τ^s. Similarly, if $z_i^b > 0$ and $l_i > 0$, then $p_i^{pro}(l_i) = p_i + \tau^b$ holds from Eqs. (2), (4b), and (4c), implying that prosumers equate their marginal benefit with the sum of p_i and the transmission tariff τ^b when purchasing the power z_i^b from the grid.

Net Metering: Under a net-metering system, there can be one meter recording the power injection ($z_i > 0$) and withdrawal ($z_i < 0$). Both types of transactions are subject to τ, which is considered a "subsidy" for sale ($z_i > 0$) or an end-user transmission tariff for purchase ($z_i < 0$). The optimization problem faced by prosumers at node i under a net-metering system is displayed as follows:

$$\underset{l_i,\, g_i \geq 0,\, z_i \text{ free}}{\text{maximize}} \quad (p_i + \tau) z_i + \int_0^{l_i} p_i^{pro}(m_i) dm_i - C_i^g(g_i) \tag{5a}$$

subject to

$$z_i + l_i - K_i - g_i \leq 0 \qquad (\delta_i), \tag{5b}$$

$$g_i \leq G_i \qquad (\kappa_i) \tag{5c}$$

The corresponding KKT conditions of the prosumers' problem in node i under the net-metering system are collected for all nodes as in (6a)–(6e).

$$p_i + \tau - \delta_i = 0, \forall i \tag{6a}$$

$$0 \leq l_i \perp P_i^0 - \frac{P_i^0}{\alpha_i Q_i^0} l_i - \delta_i \leq 0, \forall i \tag{6b}$$

$$0 \leq g_i \perp -C_i^{g\prime}(g_i) + \delta_i - \kappa_i \leq 0, \forall i \tag{6c}$$

$$0 \leq \delta_i \perp z_i + l_i - K_i - g_i \leq 0, \forall i \tag{6d}$$

$$0 \leq \kappa_i \perp g_i - G_i \leq 0, \forall i \tag{6e}$$

Eq. (5b) holds as an equality at the optimum. By substituting $z_i = K_i + g_i - l_i$ into the objective function, the maximization problem (5) can be equivalently expressed as follows:

$$\underset{l_i,\, g_i \geq 0}{\text{maximize}} \quad (p_i + \tau)(K_i + g_i) - (p_i + \tau) l_i + \int_0^{l_i} p_i^{pro}(m_i) dm_i - C_i^g(g_i) \tag{7a}$$

subject to

$$g_i \leq G_i \qquad (\kappa_i) \tag{7b}$$

This reformulation implies that under the net-metering scheme, prosumers act as a "consumer" and face the retail price $p_i + \tau$ for their gross consumption l_i, which is exactly the same as for conventional consumers. However, when prosumers act as a "producer," they receive more than the wholesale price p_i for all of their generation $K_i + g_i$ by effectively receiving a subsidy τ, which is different from the case for conventional producers.

Benchmark Tariff: This study considers a benchmark tariff, where in equilibrium, prosumers, either acting as a producer or a consumer, should be treated in the same way as conventional producers or consumers, respectively. In other words, prosumers acting as a producer should face the wholesale price p_i for their generation $K_i + g_i$. Similarly, prosumers acting as a consumer should be subject to the retail price $p_i + \tau$ for their gross consumption l_i. Any deviation from this principle would create an incentive for prosumers to amend their behavior in a way that is not aligned with other producers or consumers in the market. The optimization problem of prosumers in node i is then formulated as follows:

$$\underset{l_i,\, g_i \geq 0}{\text{maximize}} \quad p_i(K_i + g_i) - (p_i + \tau) l_i + \int_0^{l_i} p_i^{pro}(m_i) dm_i - C_i^g(g_i) \tag{8a}$$

subject to

$$g_i \leq G_i \qquad (\kappa_i) \tag{8b}$$

The amount of energy purchased from or sold to the grid can be recovered by $z_i = K_i + g_i - l_i$. The corresponding KKT conditions of the prosumers' problem in node i under the benchmark tariff are collected for all nodes as in (9a)–(9c).

$$0 \leq l_i \perp P_i^0 - \frac{P_i^0}{\alpha_i Q_i^0} l_i - (p_i + \tau) \leq 0, \forall i \tag{9a}$$

$$0 \leq g_i \perp p_i - C_i^{g\prime}(g_i) - \kappa_i \leq 0, \forall i \tag{9b}$$

$$0 \leq \kappa_i \perp g_i - G_i \leq 0, \forall i \tag{9c}$$

2.4. The independent system operator

This study assumes that the ISO collects offers from producers and bids from utilities and maximizes the wholesale social surplus (10a),

[7] It could also be other types of fixed or sunk costs that a utility attempts to recover, e.g., renewable procurement cost, those associated with discounts to low-income households, and legacy stranded costs.

[8] Prosumers who participate in a day-ahead wholesale market would face inherent uncertainty of non-dispatchable renewable power sources. In such occasions, prosumers' attitude toward risk would affect the quantities they supply and demand. Particularly, if prosumers are risk averse, they may discount or derate the renewable output achievable in the real time.

i.e., the consumer benefit at the wholesale level minus the cost for producer. The decision variables are quantities demanded, d_i, generation x_{ih} from power plant h in node i with a cost $C_{ih}(x_{ih})$, and net injection to or withdrawal from the grid, y_i.

$$\underset{d_i, x_{ih} \geq 0, y_i \text{ free}}{\text{maximize}} \quad \sum_i \int_0^{d_i} p_i(n_i)\, dn_i - \sum_{i,h\in\mathcal{H}_i} C_{ih}(x_{ih}) \tag{10a}$$

subject to

$$x_{ih} - X_{ih} \leq 0 \qquad (\rho_{ih}),\ \forall i, h \in \mathcal{H}_i, \tag{10b}$$

$$-\sum_i y_i = 0 \qquad (\theta), \tag{10c}$$

$$\sum_i PTDF_{ki} y_i \leq T_k \qquad (\lambda_k^+),\ \forall k, \tag{10d}$$

$$-\sum_i PTDF_{ki} y_i \leq T_k \qquad (\lambda_k^-),\ \forall k, \tag{10e}$$

$$y_i - \sum_{h\in\mathcal{H}_i} x_{ih} - (z_i^s - z_i^b) + d_i = 0 \qquad (p_i),\ \forall i, \tag{10f}$$

$$\text{or} \quad y_i - \sum_{h\in\mathcal{H}_i} x_{ih} - z_i + d_i = 0 \qquad (p_i),\ \forall i \tag{10g}$$

Eq. (10b) limits the output x_{ih} to be less than its capacity X_{ih}. Eq. (10c) ensures that the total net injection/withdrawal in the system is equal to zero. Eqs. (10d) and (10e) describe that flow in line k is less than its transmission limit T_k. The term $PTDF_{ki}$ represents the power transfer distribution factors based on linearized-DC power flows. The final two constraints define the nodal balance in each node with its dual variable, yielding the nodal power prices p_i. These constraints are case-dependent, that is, (10f) and (10g), representing the net-billing and the net-metering cases, respectively. The inclusion of the nodal balance (10f) or (10g) shifts the demand of conventional consumers in the wholesale market operated by the ISO, given the decision z_i of prosumers. For example, if prosumers sell $z_i > 0$ in node i, the effective "wholesale" demand is then reduced or shifts to the left by z_i through (10g). Similarly, if prosumers purchase $z_i < 0$ from the grid, the wholesale demand increases or shifts to the right by z_i, reflecting the demand from both conventional consumers and prosumers. For the benchmark case, $z_i = K_i + g_i - l_i$ is substituted in (10g).

The ISO's KKT conditions are then given as follows:

$$0 \leq d_i \perp P_i^0 - \frac{P_i^0}{(1-\alpha_i)Q_i^0} d_i - (p_i + \tau^b) \leq 0, \forall i \tag{11a}$$

$$\text{or} \quad 0 \leq d_i \perp P_i^0 - \frac{P_i^0}{(1-\alpha_i)Q_i^0} d_i - (p_i + \tau) \leq 0, \forall i \tag{11b}$$

$$0 \leq x_{ih} \perp -C'_{ih}(x_{ih}) - \rho_{ih} + p_i \leq 0, \forall i, h \in \mathcal{H}_i \tag{11c}$$

$$\theta + \sum_k PTDF_{ki}(\lambda_k^- - \lambda_k^+) - p_i = 0, \forall i \tag{11d}$$

$$0 \leq \rho_{ih} \perp x_{ih} - X_{ih} \leq 0, \forall i, h \in \mathcal{H}_i \tag{11e}$$

$$-\sum_i y_i = 0 \tag{11f}$$

$$0 \leq \lambda_k^+ \perp \sum_i PTDF_{ki} y_i - T_k \leq 0, \forall k \tag{11g}$$

$$0 \leq \lambda_k^- \perp -\sum_i PTDF_{ki} y_i - T_k \leq 0, \forall k \tag{11h}$$

$$y_i - \sum_{h\in\mathcal{H}_i} x_{ih} - (z_i^s - z_i^b) + d_i = 0, \forall i \tag{11i}$$

$$\text{or} \quad y_i - \sum_{h\in\mathcal{H}_i} x_{ih} - z_i + d_i = 0, \forall i \tag{11j}$$

Assuming interior solutions for the primal variables, Eqs. (11a) or (11b) together with Eq. (1) state that conventional consumers evaluate their marginal benefit $p_i^{con}(d_i) = P_i^0 - \frac{P_i^0}{(1-\alpha_i)Q_i^0} d_i$ against the retail price, that is, the wholesale price p_i plus the transmission charge τ^b or τ, respectively, when deciding their quantity demand, d_i.

Note that the ISO earns a congestion revenue (surplus) expressed as $\sum_i (p_i - \theta) y_i$, where θ is called the system price or hub price [24,25]. If there is no congestion in the network, the ISO obtains no surplus because the nodal price p_i equals the system price θ for all nodes.

2.5. Fixed cost recovery

While each participant's optimization problem represents its behavior in the power market, Eqs. (12)–(14) help determine the transmission charge to reimburse to the transmission owners for their revenue adequacy for net-billing, net-metering, and the benchmark cases, respectively. Specifically, F represents the revenue requirement, which accounts for the payment to or receipt from prosumers, same for all cases. For instance, when $z_i^s > 0$ and $z_i^b = 0$ under net billing in (12), the transmission owners' net receipt is equal to $d_i\tau^b - z_i^s\tau^s$, the receipt from conventional consumers minus the payment to ($\tau^s > 0$) or receipt from ($\tau^s < 0$) prosumers. Note that if R in the second equation of (12) is positive (negative), then $\tau^s > 0$ ($\tau^s < 0$). On the other hand, when $z_i^b > 0$ and $z_i^s = 0$, the net receipt by the transmission owners constitutes the receipt from both consumers and prosumers, i.e., $d_i\tau^b + z_i^b\tau^b$. Under net metering in (13) when $z_i > 0$ ($z_i < 0$), the transmission owners receive $d_i\tau$ from consumers, while $z_i\tau$ is paid to (received from) prosumers. In all cases, the transmission owners' net receipt needs to equal F.[9]

$$\sum_i \left(-z_i^s\tau^s + z_i^b\tau^b + d_i\tau^b\right) = F, \quad \tau^s = R\tau^b \tag{12}$$

$$\sum_i \left(-z_i + d_i\right)\tau = F \tag{13}$$

$$\sum_i \left(l_i + d_i\right)\tau = F \tag{14}$$

The entire problem is then solved simultaneously by the collection of KKT conditions and the condition for revenue adequacy. For the net-billing case, they constitute (4a)–(4f), (11a), (11c)–(11i), and (12). For the net-metering case, they include (6a)–(6e), (11b)–(11h), (11j), and (13). For the benchmark case, they are (9a)–(9c), (11b)–(11h), (11j), and (14).

3. Analysis

This section analytically compares the net-metering scheme with the benchmark since the former is the most commonly implemented. To highlight the essential differences between these two schemes, it bypasses some engineering and market details discussed in the previous section, specifically the backup generation of prosumers, transmission flows, and generation capacity. Along this line, it considers an entire region without the index i for the nodes. Interior solutions are also assumed for each variable.

The market equilibrium consists of three key conditions. First, (6a), (6b), and (9a) imply that prosumers face the retail price $p + \tau$ in both schemes as well as conventional consumers as alluded to in (11b). Thus, $p^{pro}(l) = p^{con}(d) = p^r(l + d) = p + \tau$. Noting that the total quantity demanded is supplied by conventional and renewable generation, that is, $l + d = x + K$, we have $p^r(x + K) = p + \tau$. Second, conventional power producers face the wholesale price p as in (11c), meaning that $C'(x) = p$ in both schemes. Third, the conditions for fixed cost recovery are different among the two systems. In the net-metering scheme, we have $(-z + d)\tau = x\tau = F$ from (13) and (11j), while the benchmark is subject to $(l + d)\tau = (x + K)\tau = F$ as in (14).

For the qualitative analysis, it is assumed that $p^r(\cdot)$ and $C(\cdot)$ are continuously differentiable, and furthermore, $p^{r\prime}(\cdot) < 0$ and $C''(\cdot) > 0$, implying that the retail demand curve and the marginal cost (or

[9] In case the cost is recovered through a combination of fixed charge and volumetric rate, then F can be viewed as a "residual" cost after netting the fixed charge component. Note that the fixed charge does not affect the prosumers' operation decisions.

supply) curve are downward- and upward-sloping, respectively, which is a standard assumption for power markets. Note that (13) and (14), or $x\tau = F$ and $(x + K)\tau = F$ are non-linear equations that include bi-linear terms. Hence, multiple equilibria may exist. In such a case, an equilibrium with the lowest transmission tariff τ is selected since it would be politically more acceptable. Using the superscripts NM and BM for the equilibrium outcome of the net-metering and benchmark systems, respectively, the following results are obtained:

Proposition 1. *In equilibrium,* $x^{BM} > x^{NM}$, $l^{BM} + d^{BM} > l^{NM} + d^{NM}$, $\tau^{BM} < \tau^{NM}$, $p^r(l^{BM} + d^{BM}) < p^r(l^{NM} + d^{NM})$, *and* $p^{BM} > p^{NM}$ *hold.*

Proof. It follows from $p^r(x + K) = p + \tau$ and $C'(x) = p$ that $p^r(x + K) - C'(x) = \tau$. For the net metering scheme, the condition for fixed cost recovery is then expressed as $x^{NM}\tau^{NM} = x^{NM}\left(p^r(x^{NM} + K) - C'(x^{NM})\right) = F$. Note that $(x^{NM} + K)\tau^{NM}$ generates more revenue than F, that is, $(x^{NM} + K)\left(p^r(x^{NM} + K) - C'(x^{NM})\right) > F$. Let x increase from x^{NM}. Then, $p^r(x + K)$ decreases, while $C'(x)$ increases, from the assumption of $p^{r\prime}(\cdot) < 0$ and $C''(\cdot) > 0$. Consequently, $p^r(x+K)-C'(x) = \tau$ decreases below the level of τ^{NM}, approaching 0 (no revenue). Since the functions are continuous, we can find $x^{BM} > x^{NM}$ and $\tau^{BM} < \tau^{NM}$ such that $(x^{BM} + K)\tau^{BM} = (x^{BM} + K)\left(p^r(x^{BM} + K) - C'(x^{BM})\right) = F$. Noting that $l + d = x + K$, $l^{BM} + d^{BM} > l^{NM} + d^{NM}$ holds from $x^{BM} > x^{NM}$. Moreover, $p^r(l^{BM} + d^{BM}) < p^r(l^{NM} + d^{NM})$ and $p^{BM} = C'(x^{BM}) > C'(x^{NM}) = p^{NM}$ hold true.

Proposition 2. *In equilibrium, conventional consumer, producer, and social surpluses either inclusive or exclusive of prosumers under the benchmark scheme are greater than those under the net-metering scheme. Prosumer surplus when selling excess power is less under the benchmark scheme than the net-metering scheme.*

Sketch of Proof. From Proposition 1, conventional consumers and producers face respectively lower retail price (less payment) and higher wholesale price (greater revenue), and hence their surpluses are greater under the benchmark scheme than under the net-metering scheme. This also holds for social surplus exclusive of prosumers, which is the sum of both conventional consumer and producer surpluses. Prosumer surplus under the benchmark scheme subtracted by that under the net-metering scheme consists of two parts, that is, $A = \int_{p^{BM}+\tau^{BM}}^{p^{NM}+\tau^{NM}} \left(p^{pro}\right)^{-1} dn - K(p^{NM} + \tau^{NM} - p^{BM} - \tau^{BM})$ and $-K\tau^{BM}$. The first part A is negative when selling excess power ($K > l$), while it is positive when purchasing power ($K < l$). The second part $-K\tau^{BM}$ is always negative. Thus, prosumer surplus under the benchmark scheme when selling excess power is less than that of the net-metering scheme. When prosumers sell power, the negative term A is canceled out by the increase in conventional consumer surplus (i.e., a wealth transfer from prosumers to conventional consumers). When prosumers buy power, the positive term A represents an increase in social surplus. On the other hand, the negative term $-K\tau^{BM}$ is merely a wealth transfer to transmission owners for revenue adequacy and thus is canceled out in terms of social surplus. In sum, the social surplus inclusive of prosumers under the benchmark scheme is greater than that under the net-metering scheme.

Remark. As alluded to in the objective function (8a), prosumers face the wholesale price for their renewable generation K without a subsidy under the benchmark scheme, which reduces their surplus by $K\tau^{BM}$ compared to the net-metering scheme. Moreover, as shown in Proposition 1, the retail price is lower under the benchmark scheme than under the net-metering scheme, i.e., $p^r(l^{BM} + d^{BM}) = p^{BM} + \tau^{BM} < p^{NM} + \tau^{NM} = p^r(l^{NM} + d^{NM})$. This is another reason for the difference in prosumer surplus between the two schemes. When prosumers sell excess power ($K > l$), $K(p^{NM} + \tau^{NM} - p^{BM} - \tau^{BM})$ is greater than $\int_{p^{BM}+\tau^{BM}}^{p^{NM}+\tau^{NM}} \left(p^{pro}\right)^{-1} dn$, which represents the change in the prosumers' gross benefit for self-consumption. Hence, A becomes negative, implying that prosumers are worse off by selling excess power under the benchmark scheme. By contrast, when prosumers buy power ($K < l$), $K(p^{NM} + \tau^{NM} - p^{BM} - \tau^{BM})$ is less than $\int_{p^{BM}+\tau^{BM}}^{p^{NM}+\tau^{NM}} \left(p^{pro}\right)^{-1} dn$, resulting in positive A. This implies that prosumers are better off by A under the benchmark scheme because of the lower retail price for purchasing power from the grid. However, the overall sign for $A - K\tau^{BM}$ is ambiguous for the case of purchase.

4. Numerical case study

To illustrate the effects of different pricing schemes, this section applies the models developed in Section 2 to a case study considering a three-node network with three firms, ten generating units, and three transmission lines.[10] The marginal cost for node A ranges between \$35/MWh and \$55/MWh. Node B has two types of generating technologies: renewables with zero variable production cost and moderately expensive units with a marginal cost of approximately \$25/MWh. Node C has units with relatively low marginal costs. In the baseline case, roughly 20% and 10% of the electricity demand at node A is met by imports from nodes B and C, respectively. Demand at node A also has the highest willingness-to-pay for electricity among three nodes, reflecting households' affordability and incentives to become prosumers. The demand parameters are calibrated by solving a cost-minimizing problem with an elasticity of −0.2.[11] The data are documented in the Appendix.

For the easiness of exposition, the analysis assumes that prosumers are only located in node A. The amortized fixed cost that needs to be recovered is equal to \$18,000 for each hour. In Section 4.1, it is assumed that prosumers face a quarter amount of volumetric transmission tariff in addition to the wholesale price, p_i, when they sell energy to the grid, that is, $\tau^s = 0.25\tau^b$ (i.e., $R = 0.25$) under the net-billing policy. It is further assumed that DERs owned by prosumers produce an output of 20 MW and 500 MW, leading prosumers to act as a net buyer and a net seller in equilibrium of the wholesale market, respectively. Section 4.2 then examines more scenarios by changing the proportion α_A of prosumers in node A from 0.05 to 0.95, while maintaining no prosumers in nodes B and C. Finally, Section 4.3 assumes that prosumers are mandated to contribute to fixed cost recovery when selling power into the grid, i.e., $\tau^s = -\tau^b$ (i.e., $R = -1$), to further investigate the impact of τ^s on the outcome.

4.1. Main results

This section reports the results in Tables 2–3 for cases with 50% prosumers in node A under 20 MW and 500 MW of renewable outputs, respectively. Each table is divided into three parts, including the results associated with prosumers, wholesale market, and the rent distribution. Columns (a)–(c) correspond to the benchmark, net-metering, and net-billing cases, respectively. Overall, the numerical results discussed below regarding the benchmark and net-metering cases are broadly consistent with the analytical findings in Section 3.

A number of observations emerge from Table 2. Prosumers, which produce 20 MW of renewable output, act as a buyer to purchase power from the main grid, that is, 387.31, 376.47, and 376.47 MWh for three cases, respectively. In fact, because prosumers are subject to the same tariff when purchasing power from the grid under the

[10] The data were previously used to examine carbon leakages under California climate change policy [26]. Here, our intention is to analyze the impacts of different pricing policies on market outcomes and distributional implications.

[11] Empirical studies suggest that the price elasticities of the demand for electricity in the U.S. range from −0.1 to −0.4, which means a 1% increase in the price leads to decreases in the quantities demanded by 0.1 to 0.4% (see, for example, Alberini and Filippini [27]).

Table 2
Results of 50% prosumers and 20 MW renewable case.

Variables\Scenarios[a]	(a)	(b)	(c)
Transmission tariff [$/MWh]	8.36	8.48	8.48
Prosumers' sale(+)/purchase(−) [MWh]	−387.31	−376.47	−376.47
Prosumers' load [MWh]	407.31	407.97	407.97
Prosumers' backup generation [MWh]	0	11.50	11.50
Prosumer surplus [$K]	35.98	36.28	36.28
Conventional demand [MWh]	1745.18	1744.94	1744.94
Conventional generation [MWh]	2132.49	2121.41	2121.41
Power price in node A [$/MWh]	47.67	47.26	47.26
Sales-weighted power price [$/MWh]	31.03	30.89	30.89
Consumer surplus [$K]	101.08	101.14	101.14
Producer surplus [$K]	15.40	15.00	15.00
ISO's revenue [$K]	1.20	1.18	1.18
Social surplus (w/o prosumers) [$K]	117.68	117.32	117.32
Total social surplus [$K]	153.66	153.60	153.60

[a](a) Benchmark, (b) Net metering, and (c) Net billing.

Table 3
Results of 50% prosumers and 500 MW renewable case.

Variables\Scenarios[a]	(a)	(b)	(c)
Transmission tariff [$/MWh]	8.15	10.84	10.39
Prosumers' sale(+)/purchase(−) [MWh]	73.29	79.99	60.21
Prosumers' load [MWh]	426.71	420.01	439.78
Prosumers' backup generation [MWh]	0	0	0
Prosumer surplus [$K]	58.28	62.55	61.99
Conventional demand [MWh]	1782.52	1740.64	1746.86
Conventional generation [MWh]	1709.23	1660.64	1686.64
Power price in node A [$/MWh]	39.68	39.52	39.71
Sales-weighted power price [$/MWh]	28.21	28.11	28.18
Consumer surplus [$K]	106.74	102.28	102.90
Producer surplus [$K]	8.64	8.38	8.52
ISO's revenue [$K]	0.84	0.84	0.84
Social surplus (w/o prosumers) [$K]	116.22	111.49	112.26
Total social surplus [$K]	174.50	174.04	174.24

[a](a) Benchmark, (b) Net metering, and (c) Net billing.

net-metering and net-billing cases, two cases collapse and produce the same market outcomes.[12] The transmission tariff of $8.36/MWh for the benchmark is lower than $8.48/MWh for net metering as in Proposition 1. The quantity demanded by prosumers in columns (b) and (c) is marginally larger than the benchmark case, that is, 407.97 versus 407.31 MWh, mainly due to positive self-generation of backup resources. Their load of 407.97 MWh in these two cases is met by dispatchable self-generation (11.50 MWh), renewables (20 MWh), and the supply from the wholesale market (376.47 MWh). In the wholesale market, the conventional demand and generation under the benchmark are greater than those in columns (b) and (c). The wholesale power price in node A under the benchmark is higher than that in columns (b) and (c) by $0.4/MWh (= 47.67 – 47.26) as prosumers buy an additional 10.8 MWh of power from the grid. This also results in a higher sales-weighted wholesale price by $0.14/MWh under the benchmark case.

Next, as shown in Proposition 2, the results of economic rent distribution indicate that the benchmark has the highest value regarding both the "total social surplus" inclusive of prosumers and the wholesale social surplus exclusive of prosumers. Overall, the net-billing and net-metering policies benefit prosumers by $0.3 K compared to the benchmark case. By contrast, conventional producers experience a surplus decline by $0.4 K in (b) and (c), suggesting that net-billing and net-metering policies benefit prosumers at the expense of producers.[13]

This section next turns to the case in which prosumers produce 500 MW of renewable output in Table 3. The transmission tariff remains the lowest under the benchmark. The abundance of renewables allows prosumers to act as a net seller in equilibrium, selling 73.29, 79.99, and 60.21 MWh of energy into the grid for the three cases, respectively. This also leads to a divergence of market outcomes in (b) and (c), where, relative to the net-metering case, the net-billing policy discourages prosumers from selling excess power into the grid because they face less subsidy for sales (i.e., $\tau^s = 0.25\tau^b$) under net billing. More specifically, column (c) suggests that under the net-billing case, prosumers sell less power into the grid compared to the benchmark and the net-metering case by 18% and 25%, respectively. Correspondingly, prosumers under the net-billing case decide to self-consume their renewable output more, i.e., 13.07 MWh (= 439.78 – 426.71) and 19.77 MWh (= 439.78 – 420.01), respectively. The power price in node A is inversely related to the amount of energy sold by prosumers: i.e., the least amount leads to the highest price in the net-billing case.

With regard to the economic rent distribution, the prosumer surplus is the worst (best) in the benchmark (net-metering) case. The producer surplus is directly related to the sales-weighted power price. The higher it is, the larger the producer surplus is as seen in the benchmark. Both producer and consumer surpluses show a reverse order of the prosumer surplus when comparing among the three cases. Thus, the net-metering policy, while encouraging prosumers selling more power into the grid, results in more wealth transfer from conventional producers and consumers when compared to the net-billing policy. As shown in Table 3, net metering leads to a decline of surpluses from the benchmark by $4.46 K (= 106.74 – 102.28) and $0.26 K (= 8.64 – 8.38) for consumers and producers, respectively, compared to that under the net-billing case, which is respectively $3.84 K (= 106.74 – 102.90) and $0.12 K (= 8.64 – 8.52). Overall, the social surplus, both inclusive or exclusive of prosumers, is the highest for the benchmark case.

Finally, it is noted that the numeric results are consistent with the findings in Section 3 with two exceptions: i.e., total consumption and consumer surplus in net metering are slightly greater than those under the benchmark in Table 2. The reason is that prosumers' dispatchable self-generation, which is not considered in Section 3, leads to more consumption by prosumers in column (b) than in (a). Moreover, the increase in the sum of dispatchable self-generation and conventional generation pushes down the retail price, and hence increases the consumer surplus in (b).

4.2. Impacts of the size of prosumers

Whether prosumers are a buyer or a seller is also affected by their size in node A as well as the amount of renewables. This section reports the results of an analysis that varies the proportion of prosumers in node A from $\alpha_A = 0.05$ to $\alpha_A = 0.95$ when the renewable output is equal to 500 MW. The interest is on the changes in prosumers' net position and their impact on the economic rent distribution.

Fig. 2(a) displays the sale (+) purchase (−) decision by prosumers in equilibrium. Note that the net-billing and net-metering cases give identical solutions when prosumers are a buyer or when their presence in node A is more than $\alpha_A = 0.6$. Although not discernible, the net-billing case represents the lower bound of transactions (i.e., sell less) when prosumers have excess energy. This is a direct consequence of the policy, $\tau^s = 0.25\tau^b$, which discourages selling excess power into the grid. This result is also consistent with Table 3. Although it is also less discernible, the benchmark case forms the lower envelope (i.e. buying more) when prosumers act as a buyer withdrawing energy from the

[12] This observation can also be validated through their equivalent KKT conditions when $z_i^b > 0$ and $z_i^s = 0$ under the net-billing case. Specifically, Eqs. (4b) and (6a) become identical under this situation.

[13] Empirically, this is evident by the court's recent rejection of Ameren's request to abandon full retail net metering for new home solar panels in Illinois and highlights the concerns of revenue erosion under the net-metering policy [28].

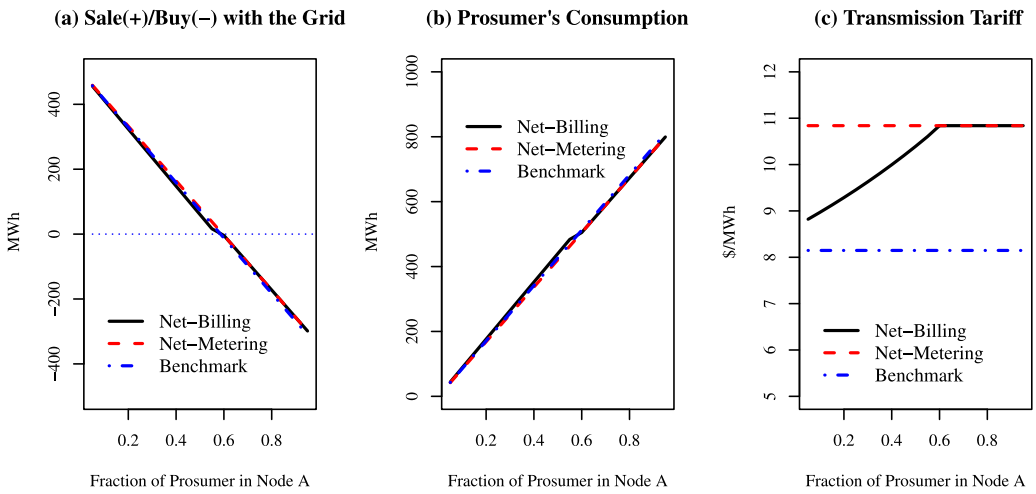


Fig. 2. Plots of prosumers' transactions with the main grid, consumption, and volumetric tariff against the proportion of prosumers in node A under $\tau^s = 0.25\tau^b$.

wholesale market. Consequently, it also delineates the upper envelope in Fig. 2(b) that represents prosumers' consumption.

As discussed in Section 2, both prosumers and consumers face the same retail price $p_i + \tau$ under the net-metering and benchmark cases. Thus, the aggregate demand of prosumers and consumers is determined by the horizontally aggregated retail inverse demand function, $p_i^r(d_i + l_i) = P_i^0 - \frac{P_i^0}{Q_i^0}(d_i + l_i)$, irrespective of α_i. In other words, the change in energy consumed by prosumers is offset by an opposite change in consumption by consumers at the corresponding retail price when changing α_i, leaving the total demand of consumers and prosumers unchanged. As a result, both net-metering and benchmark cases exhibit constant transmission tariffs as in Fig. 2(c). The tariff under the net-metering case is higher than that of the benchmark case by \$2.7/MWh or 33%, as in Proposition 1. On the other hand, in the net-billing case, prosumers and consumers do not face the same retail price when prosumers sell power because of the construction of tariff, $\tau^s = 0.25\tau^b$. This implies that the transmission tariff is not constant when varying α_i. Indeed, the transmission tariff continues to rise in commensurate with α_A in the net-billing case until $\alpha_A = 0.6$, and it then reaches plateaus when prosumers buy electricity, collapsing to the level of the net-metering case. The reason is that $\tau^s < \tau^b$ discourages sales to the grid, enticing prosumers to self-consume more. Prosumers increase their self-consumption with α_A, and hence contribute less to the fixed cost recovery, thereby driving up the transmission tariff.

Fig. 3 illustrates the impacts on the economic rent distribution. The outcome is affected by the prosumers' decision about how much to buy from or sell into the main grid while subject to different pricing rules. As alluded to earlier, under the net-metering case, the total demand remains unchanged regardless of α_A, as is the wholesale power price as well as the producer surplus in Fig. 3(a). However, the fact that the total demand and wholesale power price are both higher under the benchmark case (see also Proposition 1 and Table 3) indicates that the producer surplus is also higher, in accordance with Proposition 2. On the other hand, the producer surplus under the net-billing case continues to decline until $\alpha_A = 0.6$, and converges to the net-metering case. Although the increase in self-consumption by prosumers offers producers opportunities to expand their sales to conventional consumers, the continuous increase in the transmission tariff shown in Fig. 2(c) and, hence, the increase in the retail power prices discourages conventional consumers from consuming power, thus reducing producers' revenue and making them worse off. Turning to the consumer surplus in Fig. 3(b), it declines with α_A in all cases as more consumers convert to prosumers with increasing prosumer surplus. The consumer surplus in the benchmark case is the highest among the three cases, benefiting from lower transmission tariffs and retail prices.

The ISO's revenue deserves some attention. Under the benchmark and net-metering cases, consumers and prosumers in node A are subject to the same retail price regardless of α_A. Thus, any decision by prosumers to increase or decrease their consumption (that leads to a decline or an increase of sales to the grid) is entirely offset by the actions of consumers in node A. This results in the same injection/withdrawal patterns, and so is the ISO's revenue. For the net-billing scheme, the same observation only applies to the case where prosumers act as a buyer since the outcomes under net-billing case collapse to the net-metering case when they are in a short position. On the other hand, a decline of sales into the grid by prosumers under a long position entices producers elsewhere to export energy to node A and elsewhere in the network, thereby increasing transmission flows and consequently elevating the ISO's revenue. As the sales by prosumers drop with α_A, the ISO's revenue also increases accordingly in Fig. 3(c). The fact that the prosumers' net position alters from net selling to net buying also creates a kink in the ISO's surplus.

The prosumer surplus in Fig. 3(d) increases in all cases when there are more prosumers in node A with more consumers converting to prosumers. Prosumers benefit greatly from the net-metering policy, forming the upper envelope irrespective of the value of α. The impacts on the surplus in the wholesale market are shown in Fig. 3(e). Essentially, the wholesale social surplus (excluding prosumer surplus) worsens when prosumers increase their presence in node A. Of course, this decline is in part due to the forgone surplus associated with consumers who convert to prosumers. Finally, the total social surplus (inclusive of prosumer surplus) in Fig. 3(f) is a joint effect shown in Figs. 3(a)–(e). In the net-billing case, the decline in the wholesale social surplus when $\alpha_A \leq 0.6$ is not sufficiently compensated for by the increased surplus earned by prosumers in Fig. 3(e), leading to a decline in the total social surplus until it collapses to the net-metering case. The result in Fig. 3(f) is also consistent with Proposition 2 in that the social surpluses inclusive or exclusive of prosumers under the benchmark case are greater than those under the net-metering case.

4.3. Sensitivity analysis

This section further extends the setup of Section 4.2 to the case for $\tau^s = -\tau^b$, where prosumers face less than the wholesale price p_i when selling power to the grid. A number of observations different from the previous cases emerge in Figs. 4–5. First, the revenue that can be earned from the sales by prosumers in this situation is less, thereby discouraging them to sell power into the grid, evident in Fig. 4(a), while increasing self-consumption as shown in Fig. 4(b). Second, when prosumers shoulder the burden of recovering fixed costs under $\alpha_A \leq 0.3$ with relatively large amount of sales, the transmission tariff in Fig. 4(c)

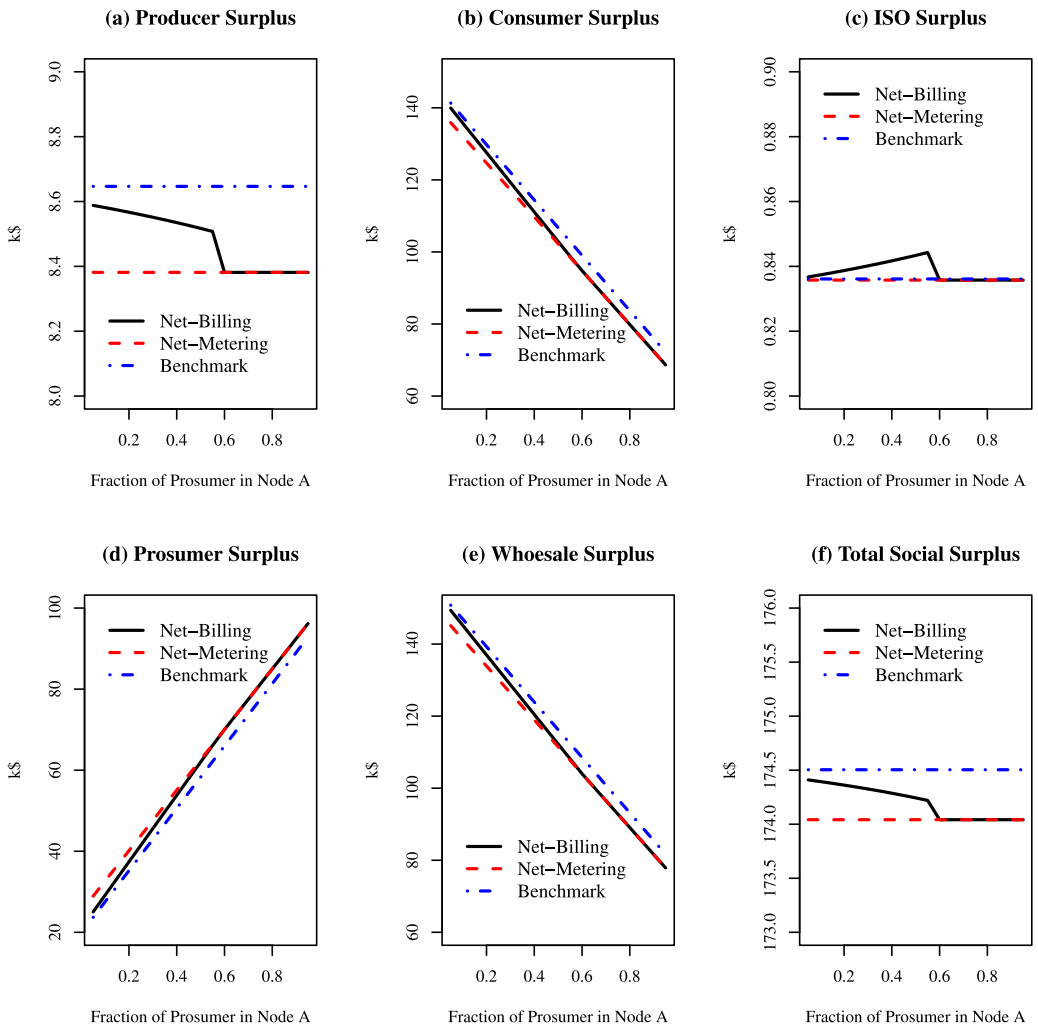


Fig. 3. Plots of the economic rent distribution against the proportion of prosumers in node A under $\tau^s = 0.25\tau^b$.

can be lower than that in the benchmark case, which decreases retail prices and raises the wholesale social surplus as indicated in Fig. 5(e). Third, to the authors' surprise, the fact that the total social surplus in Fig. 5(f) is the highest for the net-billing case, even greater than the benchmark, when $\alpha_A \leq 0.3$ suggests that requiring prosumers to contribute to fixed cost recovery when they sell power into the grid can be welfare enhancing if they are in a relatively long position. This observation is corroborated by the lower transmission tariffs in Fig. 4(c) in the net-billing case, which leads to the lower retail rates and the higher consumer surplus in Fig. 5(b), along with the higher producer surplus in Fig. 5(a) due to larger quantities demanded by consumers in the wholesale market. These jointly result in the higher wholesale social surplus in the net-billing case reported in Fig. 5(e) when α_A is less than 0.3. The lower prosumer surplus in Fig. 5(d) under the net-billing scheme compared to other two cases is more than compensated by the increases in the wholesale social surplus, resulting in the higher total social surplus in Fig. 5(f).

5. Conclusions

The lumpiness of an investment, a type of nonconvex cost, has historically presented a significant regulatory challenge for utilities in the power sector trying to recover their costs. The situation is complicated by the growing presence of prosumers when the sector is in pursuit of resilience and sustainability. The self-interest of prosumers may not align with the objective of the wholesale market. Furthermore, ill-designed policies create distortions and affect both the allocative efficiency and wealth distribution.

This study examines how prosumers affect the market equilibria when they are subject to the net-metering and net-billing policies, and further compares those outcomes to the benchmark case where prosumers are subject to the same price signals as producers (consumers) when selling (purchasing) power into (from) the grid. While FERC Order 2222 creates opportunities that enable the industry to integrate a torrent of small-size prosumers to participate in the market, it also introduces new regulatory challenges.

Overall, the results indicate that neither a net-metering nor a net-billing policy is likely to be efficient in many instances. The distortion created by the transmission tariff under such schemes may introduce further inefficiency. Prosumers can be economically better off under the net-metering system at the expense of producers and consumers in comparison to the net-billing scheme. However, to the authors' surprise, mandating prosumers to contribute to or to shoulder the burden of fixed cost recovery when they are in a relatively long position and sell a significant amount of power into the grid under the net-billing scheme can be welfare-improving. Particularly, the effect of lowering the transmission tariff can be significant enough to increase the total social surplus. Thus, the findings constitute a guiding principle to policy-makers. That is, a welfare-improving tariff entails fixed cost recovery by charging/compensating prosumers differently depending on their net position.

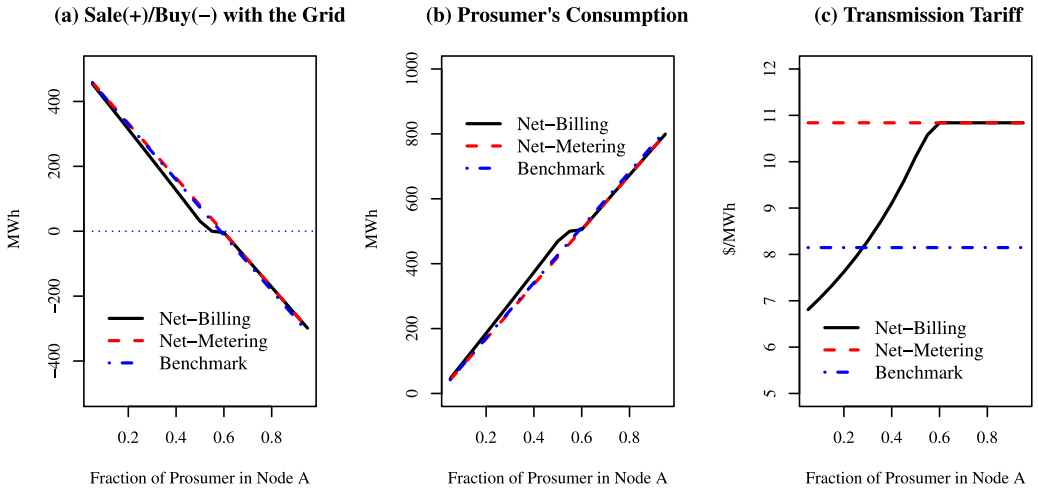


Fig. 4. Plots of prosumers' transactions with the main grid, consumption, and volumetric tariff against the proportion of prosumers in node A under $\tau^s = -\tau^b$.

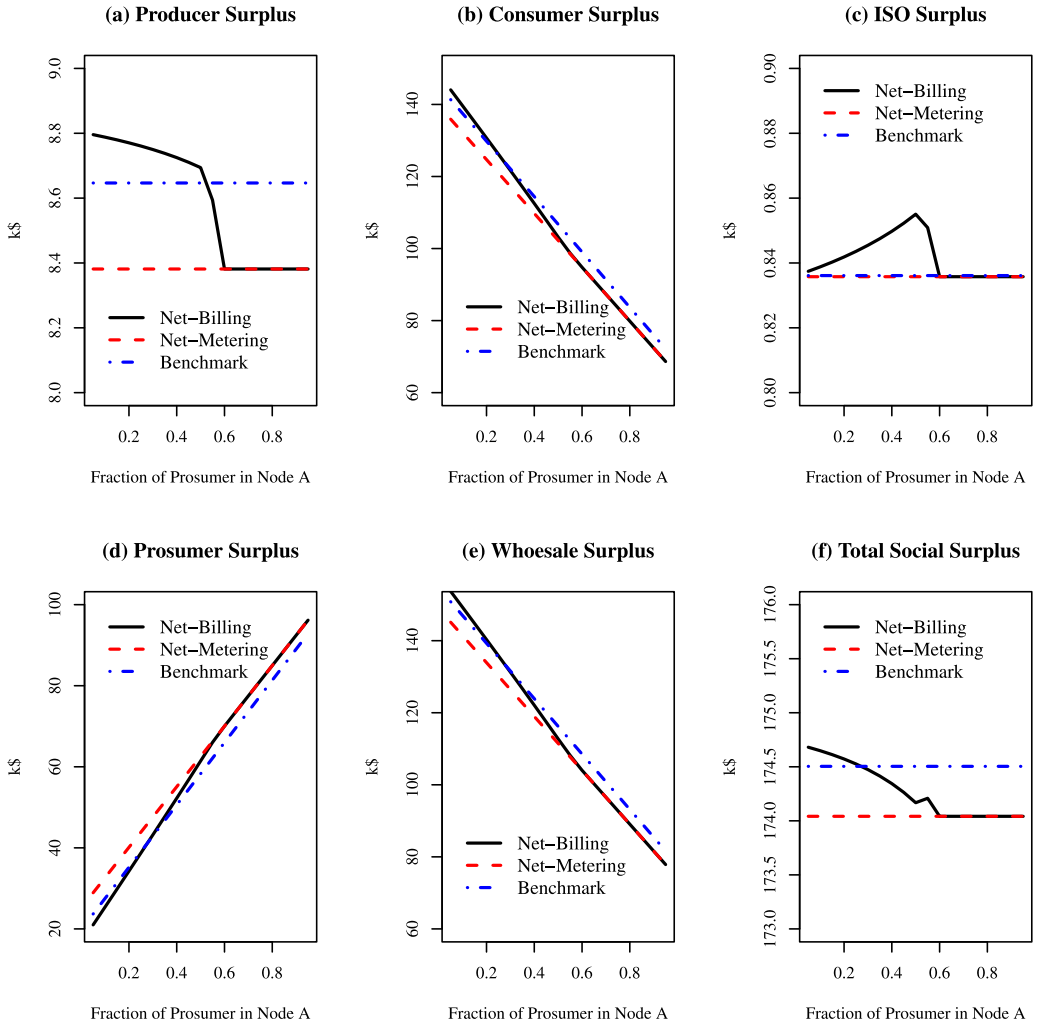


Fig. 5. Plots of the economic rent distribution against the proportion of prosumers in node A under $\tau^s = -\tau^b$.

Table A.1
Generator data of the three-node example.

Plant	Intercept ($/MWh)	Slope ($/MWh^2)	Capacity (MW)	Node (unitless)
1	38	0.02	250	A
2	35.72	0.03	200	A
3	36.8	0.04	450	A
4	15.52	0.01	150	B
5	16.2	0.02	200	B
6	0	0.001	200	B
7	17.6	0.02	400	C
8	16.64	0.01	400	C
9	19.4	0.01	450	C
10	18.6	0.02	200	C
11[a]	50	0.5	50	A

[a]Back up generation owned by prosumers at node A.

Table A.2
Intercepts of inverse demand functions of the three-node example.

Node	P_i^0 ($/MWh)	Q_i^0 (MWh)
A	228	1080
B	170	660
C	112	1146

While this study simulates different levels of renewable outputs, the analysis is essentially deterministic. Future work that incorporates uncertainty of renewable outputs into the framework using, for example, stochastic programming and robust optimization would provide a more realistic representation of power markets. Future research may also explore a variety of additional analyses, such as investments in renewable energy sources and implementation of other pricing schemes.

CRediT authorship contribution statement

Yihsu Chen: Conception/design, Material preparation, Data collection and analysis, Writing – original draft. **Makoto Tanaka:** Conception/design, Material preparation, Data collection and analysis, Writing – original draft. **Ryuta Takashima:** Conception/design, Material preparation, Data collection and analysis, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yihsu Chen reports financial support was provided by National Science Foundation.

Data availability

The data is provided in the appendix.

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All authors approved the version of the manuscript to be published.

Appendix. Data

This appendix documents the data used in the analysis, particularly regarding the generators and demand in Tables A.1–A.2, respectively [26]. Linear marginal costs and linear demand are assumed. Table A.1 summarizes the characteristics of the generators at three nodes. Node A is designed to resemble a California's power system mainly comprising natural gas plants. Node B, representing a northwest state, includes hydropower with almost zero variable production cost, and node C represents a southwest state primarily consisting of coal plants with relatively low production costs. The data regarding the demand are shown in Table A.2, in which Node A has large vertical and horizontal intercepts, corresponding to the assumption of high power demand as in California. By contrast, nodes B and C exhibit moderate electricity demand for the northwest and southwest states.

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